

Silas Kwabla Gah

PhD Candidate in Computer Science

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Computer Vision · Foundation Models · Multimodal AI · Reinforcement Learning

Research Profile

I am a PhD Candidate in Computer Science at the University of Ghana conducting research in computer vision, foundation models, multimodal learning, and reinforcement learning. My doctoral research develops a unified **Inference-Time Coordination** framework that enables frozen pretrained foundation models to adapt through coordinated reasoning rather than task-specific parameter optimisation.

My work spans generalized few-shot semantic segmentation, three-dimensional scene understanding, open-vocabulary visual perception, and multimodal reasoning. My long-term research vision is to develop computational frameworks that enable foundation models to coordinate, adapt, and reason efficiently under limited supervision while remaining computationally scalable and scientifically interpretable.

Research Interests

Research Interests

Computer Vision; Foundation Models; Inference-Time Learning; Multimodal Learning; Reinforcement Learning; Vision-Language Models; Visual Reasoning; Open-Vocabulary Perception; 3D Scene Understanding; Efficient Artificial Intelligence.

Education

2022–Present **PhD in Computer Science**, *University of Ghana*, Accra, Ghana

Thesis: Inference-Time Coordination of Foundation-Model Representations for Robust Visual Perception under Limited Supervision.

Supervisor: Prof. Ebenezer Owusu.

Research Programme

Current Inference-Time Coordination of Foundation Models for Robust Visual Perception under Limited Supervision.

Emerging Reinforcement Learning for Multimodal Reasoning using Process-Level Reward Design and Visual Grounding Verification.

Long-Term Adaptive Foundation Models capable of efficient perception, reasoning, and decision making without extensive task-specific retraining.

Research Contributions

Development of a unified **Inference-Time Coordination Framework** for adapting pretrained foundation-model representations without parameter optimisation.

Feature-space coordination through Test-Time Iterative Prototype Mining (TIPM) for Generalized Few-Shot Semantic Segmentation.

Geometric coordination through Cross-View Semantic Consensus for Generalized Few-Shot 3D Point Cloud Segmentation.

Semantic coordination through Training-Free Open-Vocabulary Generalized Few-Shot Semantic Segmentation.

Current research on Reinforcement Learning for Multimodal Foundation Models with process-level reward design and efficient post-training.

Publications and Research Outputs

Peer-Reviewed Publications

- 2023 Appati, J. K., Boante, L. M., Owusu, E., and **Gah, S. K.** "Transfer Learning-Based Encoder–Decoder Model for Skin Lesion Segmentation." *International Conference on Innovations in Computer Science and Engineering*, Communications in Computer and Information Science, Vol. 1841, pp. 117–128, Springer.
- 2017 **Gah, S. K.**, Gyamfi, N. K., and Katsriku, F. "Sentiment Analysis of Twitter Feeds Using Machine Learning." *Communications on Applied Electronics*, Vol. 6, No. 9.

Manuscripts Under Review or Submitted

- 2026 **Gah, S. K.** et al. "Training-Free Generalized Few-Shot Segmentation through Open-Vocabulary Semantic Arbitration." Submitted to *WACV 2027*.
- 2026 **Gah, S. K.** et al. "Cross-View Semantic Consensus for Generalized Few-Shot Three-Dimensional Point-Cloud Segmentation." Submitted to *ACCV 2026*.
- 2026 **Gah, S. K.** et al. "Test-Time Iterative Prototype Mining for Generalized Few-Shot Semantic Segmentation." Revised manuscript submitted to *SACAIR 2026*.

Research in Progress

- 2026 **Gah, S. K.** et al. "Process-Level Reward Design and Visual Grounding Verification for Multimodal Reinforcement Learning." Manuscript and experimental validation in progress.

Research Experience

- 2022–Present **Doctoral Researcher**, *Department of Computer Science, University of Ghana, Accra, Ghana*
- Developed a unified mathematical framework for inference-time coordination of frozen foundation-model representations.
 - Designed methods for generalized few-shot semantic segmentation, open-vocabulary perception, and three-dimensional point-cloud segmentation.
 - Conducted large-scale experiments using PASCAL-5ⁱ, COCO-20ⁱ, ADE-based open-world subsets, and ScanNet++.
 - Investigated CLIP, DINOv2, SAM-family models, Mask2Former, RegionPLC, and vision–language representations.
 - Performed quantitative evaluation, ablation analysis, spatial-alignment diagnosis, and qualitative visualization.
 - Developed reproducible research pipelines using PyTorch, Hugging Face, Open3D, Docker, and GPU computing.
- 2026–Present **Researcher, Multimodal Reinforcement Learning**, *Independent Research Programme*
- Developing process-level reward functions for multimodal visual reasoning and grounding.
 - Evaluating grounding rewards, exact-match rewards, soft semantic matching, and reward–outcome correlation.
 - Investigating Group Relative Policy Optimization and VERL-based post-training for vision–language models.
 - Building an experimental pipeline around Qwen2.5-VL, RegionPLC, segmentation priors, and grounding-based verification.

Selected Research Projects

- TIPM-Seg** Inference-time prototype coordination for generalized few-shot semantic segmentation. Replaces base-trained prototype refinement with non-parametric attention and test-time adaptation.
- Cross-View Consensus** Training-free geometric coordination of dense three-dimensional priors and sparse image-derived semantic observations through calibrated multi-view projection and probabilistic consensus.
- OpenV-GFSS** Training-free open-vocabulary generalized few-shot segmentation through coordination of promptable segmentation proposals, vision–language embeddings, and few-shot support centroids.

Multimodal RLVR	Process-level reward design for multimodal reasoning, combining answer correctness, semantic matching, format validity, and visual-grounding verification.
Local AI Systems Laboratory	Experimental infrastructure for local language models, multimodal inference, agent evaluation, KV-cache analysis, and benchmarking using vLLM and Qwen-family models.

Research Methods and Technical Expertise

Machine Learning	PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, MLflow, OpenCV, NumPy, Pandas.
Foundation Models	CLIP, DINOv2, SAM-family models, Mask2Former, Qwen2.5-VL, vision-language encoders, promptable segmentation.
Reinforcement Learning	GRPO, RLVR, VERL, reward modelling, rollout analysis, process-level verification, policy evaluation.
3D Vision	Point-cloud segmentation, multi-view projection, calibrated camera geometry, semantic label lifting, Open3D.
Research Computing	Linux, CUDA environments, mixed-precision inference, L4/A100/T4 GPU workflows, vLLM, Lightning AI.
Reproducibility	Git, GitHub, Docker, Kubernetes, experiment configuration, benchmark automation, quantitative and qualitative evaluation.
Programming	Python, C#, Java, JavaScript/TypeScript, SQL, C++, Swift, Kotlin.

Teaching and Research Mentoring

2020–2024	Assistant Lecturer , <i>Ghana Institute of Management and Public Administration</i> , Accra, Ghana ○ Taught undergraduate and graduate courses in Machine Learning, Computer Vision, Deep Learning, Data Structures and Algorithms, Database Systems, Software Engineering, Web Technology, and Object-Oriented Programming. ○ Supervised and assessed more than 55 student projects spanning machine learning, computer vision, databases, web applications, and software engineering.
2025–2026	Part-Time Lecturer , <i>Accra Technical University</i> , Accra, Ghana ○ Delivered Data Structures and Algorithms and Programming Fundamentals to undergraduate students.
2020–Present	Graduate Assistant and Research Mentor , <i>University of Ghana</i> , Accra, Ghana ○ Delivered guest lectures in artificial intelligence, machine learning, and computer vision. ○ Mentored graduate students in research design, implementation, experimental evaluation, and code review.

Selected Professional Experience

2025–Present	Lead Software Engineer , <i>National Ambulance Service Digital Platform</i> , Ghana ○ Led development of offline-first Android and iOS applications, backend services, and a dispatch dashboard for emergency medical operations. ○ Implemented secure APIs, synchronization workflows, authentication, encryption, database integration, and containerized deployment.
2020–Present	Managing Partner and Lead Engineer , <i>EED Soft Consult Ltd</i> , Accra, Ghana ○ Designed and delivered enterprise, educational, legal-technology, payroll, and payment-integration systems. ○ Built production machine-learning and retrieval systems, including document search, conversational AI, and RAG pipelines.

Certifications

2024	AWS Certified Cloud Practitioner, Amazon Web Services.
2024	Microsoft Azure Solutions Architecture training/certification.

Academic References

Prof. Ebenezer Owusu, *Department of Computer Science, University of Ghana*

PhD Supervisor

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Dr. Appati, *University of Ghana*

Assistant Supervisor

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Dr. Rose Atuah, *Colorado State University–Pueblo, Colorado, USA*

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